Mara Attia

Postdoctoral Researcher in Exoplanet Science

they/them • Website • ADS • ORCID • GitHub

Mara Attia obtained their **master's degree** from **École Polytechnique** (Paris) in September 2019, after a research internship on exoplanetary atmospheres and orbital architectures at the **Geneva Observatory** with Dr. Vincent Bourrier and Prof. David Ehrenreich, rewarded by the Research Internship Award in Astrophysics and Cosmology. A **second master's degree** followed at **ETH Zurich** with high honors (February 2021, 5.8/6), with a thesis under Prof. Romain Teyssier on primordial magnetic fields during reionization. They returned to Geneva in March 2021 for a **PhD** with Dr. Vincent Bourrier and Prof. Emeline Bolmont on dynamical and atmospheric evolution in close-in exoplanets, defended in 2024 with high honors after only three years and awarded the Edith Alice Müller Award.

Mara currently holds a **Postdoc.Mobility fellowship** from the Swiss National Science Foundation, begun in May 2025 at the **Kapteyn Astronomical Institute** (Groningen) with Prof. Tim Lichtenberg. Their work now centers on terrestrial and sub-Neptune exoplanets and how their interiors and atmospheres couple over a planet's lifetime, in step with the exoplanet missions of the coming years. They developed PALEOS, an open-source library of multiphase equations of state and mass–radius relations for planetary interiors, and connect it to the PROTEUS framework to follow magma oceans, atmospheric escape, and the formation of volatile-rich envelopes and sub-Neptune oceans. Several of these strands are pursued with the students they co-supervise.

AREAS OF STUDY

- Interior structure and equations of state
- Coupled interior–atmosphere evolution
- Atmospheric escape and evolution
- Magma oceans and interior melting
- Terrestrial and sub-Neptune planets
- Secular dynamical evolution

KEY NUMBERS (2021–2026)

- Published articles: 29 (7 first-author)
- Conference and seminar contributions: 27 (15 conferences, 12 seminars)
- Students supervised: 10 (5 BSc, 5 MSc)
- Open-source codes: 5
- Observing programs as PI: 4 (13.3 nights)

EDUCATION

- 2021–2024 • PhD in Astronomy and Astrophysics • University of Geneva (Geneva, Switzerland)
- 2019–2021 • MSc in Physics (Astrophysics) • ETH Zurich (Zurich, Switzerland)
- 2016–2019 • MSc in Science and Engineering • École Polytechnique (Paris, France)

POSITIONS

- 2025–present • SNSF Postdoc.Mobility Fellow • Kapteyn Astronomical Institute, University of Groningen (Groningen, Netherlands)
- 2024–2025 • Postdoctoral Researcher • Geneva Observatory, University of Geneva (Geneva, Switzerland)

AWARDS/GRANTS

- 2025 July • Edith Alice Müller Award • Swiss Society for Astronomy and Astrophysics
- 2025 May • **Postdoc.Mobility Fellowship** • Swiss National Science Foundation
- 2022 March • Travel & Funding Award • Swiss Society for Astronomy and Astrophysics

PUBLISHED ARTICLES

29 peer-reviewed articles and preprints, 7 as first author.

29. Lichtenberg T., **Attia M.**, Nicholls H., Sastre M., Bower D. J., Stuitje K., Pascal F. C., Soucasse L., Apai D., Bos P. et al. Super-Earth interiors shrink by about 10% as they crystallise. Preprint, arXiv:2609.23091 (2026).

28. **Attia M.**, Lichtenberg T. Atmospheric escape fractionates secondary but not primary atmospheres. Preprint, arXiv:2608.30106 (2026).

27. Postolec E., Lichtenberg T., Teske J. K., Nicholls H., **Attia M.**, Piette A., Dang L., Wallack N. L., Plotnykov M., McGinty A. et al. Evolutionary pathways toward survival of a thick CO_2- or SO_2-rich atmosphere on the lava world TOI-561 b. Preprint, arXiv:2609.03144 (2026).

26. **Attia M.**, Lichtenberg T., Jungová E., Sastre M. PALEOS: Multiphase equations of state and mass–radius relations for exoplanet interiors. *Astronomy & Astrophysics* 713, A225 (2026).

25. Cesario L., Lichtenberg T., **Attia M.**, Nicholls H., Kisvárdai I., Changeat Q. Reflation: redox-driven atmospheric inflation as tracer of super-Earth geochemistry. *The Astrophysical Journal Letters* 1007, L8 (2026).

24. Kahle K. A., Mollière P., Kreidberg L., Bitsch B., **Attia M.**, Dainese S. S., Storm N., Valentine D., Evans-Soma T. M., Hoeijmakers H. J. et al. The panchromatic JWST dayside spectrum of WASP-121 b reveals a refractory-rich formation. Preprint, arXiv:2606.21855 (2026).

23. **Attia M.**, Bourrier V., Bolmont E., Mignon L., Delisle J.-B., Beust H., Hara N. C., Mordasini C. The JADE code: II. Modeling the coupled orbital and atmospheric evolution of GJ 436 b to constrain its migration and companion. *Astronomy & Astrophysics* 702, A132 (2025).

22. Bourrier V., Steiner M., Castro-González A., Armstrong D. J., **Attia M.**, Gill S., Timmermans M., Fernandez J., Hawthorn F., Triaud A. H. M. J. et al. ATREIDES: I. Embarking on a trek across the exo-Neptunian landscape with the TOI-421 system. *Astronomy & Astrophysics* 701, A190 (2025).

21. **Attia M.**, Sibony Y. Exploring the parameter space of hierarchical triple black hole systems. *Astronomy & Astrophysics* 700, A35 (2025).

20. Fontanet E., Udry S., Ségransan D., Figueira P., Acevedo Barroso J. A., Akinsanmi B., **Attia M.**, Battley M., Bhatnagar S., Bugatti M. et al. CORALIE radial-velocity search for companions around evolved stars (CASCADES): IV. New planetary systems around HD 87816, HD 94890, and HD 102888 and an update on HD 121056. *Astronomy & Astrophysics* 699, A38 (2025).

19. Rodrigues J., Barros S. C. C., Santos N. C., Davoult J., **Attia M.**, Castro-González A., Sousa S. G., Demangeon O. D. S., Hobson M. J., Bossini D. et al. TOI-512: Super-Earth transiting a K-type star discovered by TESS and ESPRESSO. *Astronomy & Astrophysics* 695, A237 (2025).

18. Bourrier V., Delisle J.-B., Lovis C., Cegla H. M., Cretignier M., Allart R., Al Moulla K., Tavella S., **Attia M.**, Mounzer D. et al. The ANTARESS workflow: I. Optimal extraction of spatially resolved stellar spectra with high-resolution transit spectroscopy. *Astronomy & Astrophysics* 691, A113 (2024).

17. Psaridi A., Osborn H., Bouchy F., Lendl M., Parc L., Billot N., Broeg C., Sousa S. G., Adibekyan V., **Attia M.** et al. Discovery of two warm mini-Neptunes with contrasting densities orbiting the young K3V star TOI-815. *Astronomy & Astrophysics* 685, A5 (2024).

16. Heidari N., Boisse I., Hara N. C., Wilson T. G., Kiefer F., Hébrard G., Philipot F., Hoyer S., Stassun K. G., Henry G. W. et al. The SOPHIE search for northern extrasolar planets: XIX. A system including a cold sub-Neptune potentially transiting a V = 6.5 star HD 88986. *Astronomy & Astrophysics* 681, A55 (2024).

15. Hagelberg J., Nielsen L. D., **Attia M.**, Bourrier V., Pearce L., Venturini J., Winn J. N., Bouchy F., Bouma L. G., Briceño C. et al. TOI-858 B b: A hot Jupiter on a polar orbit in a loose binary. *Astronomy & Astrophysics* 679, A70 (2023).

14. Guilluy G., Bourrier V., Jaziri Y., Dethier W., Mounzer D., Giacobbe P., **Attia M.**, Allart R., Bonomo A. S., Dos Santos L. A. et al. DREAM: III. A helium survey in exoplanets on the edge of the hot Neptune desert with GIANO-B at TNG. *Astronomy & Astrophysics* 676, A130 (2023).

13. Psaridi A., Bouchy F., Lendl M., Akinsanmi B., Stassun K. G., Smalley B., Armstrong D. J., Howard S., Ulmer-Moll S., Grieves N. et al. Three Saturn-mass planets transiting F-type stars revealed with TESS and HARPS: TOI-615b, TOI-622b, and TOI-2641b. *Astronomy & Astrophysics* 675, A39 (2023).

12. **Attia M.**, Bourrier V., Delisle J.-B., Eggenberger P. DREAM: II. The spin–orbit angle distribution of close-in exoplanets under the lens of tides. *Astronomy & Astrophysics* 674, A120 (2023).

11. Steiner M., **Attia M.**, Ehrenreich D., Lendl M., Bourrier V., Lovis C., Seidel J. V., Sousa S. G., Mounzer D., Astudillo-Defru N. et al. Hot Exoplanet Atmospheres Resolved with Transit Spectroscopy (HEARTS): VIII. Nondetection of sodium in the atmosphere of the aligned planet KELT-10b. *Astronomy & Astrophysics* 672, A134 (2023).

10. Bourrier V., **Attia M.**, Mallonn M., Marret A., Lendl M., Konig P.-C., Krenn A., Cretignier M., Allart R., Henry G. et al. DREAM: I. Orbital architecture orrery. *Astronomy & Astrophysics* 669, A63 (2023).

9. Mounzer D., Lovis C., Seidel J. V., **Attia M.**, Allart R., Bourrier V., Ehrenreich D., Wyttenbach A., Astudillo-Defru N., Beatty T. G. et al. Hot Exoplanet Atmospheres Resolved with Transit Spectroscopy (HEARTS): VII. Detection of sodium on the long-transiting inflated sub-Saturn KELT-11 b. *Astronomy & Astrophysics* 668, A1 (2022).

8. Grouffal S., Santerne A., Bourrier V., Dumusque X., Triaud A. H. M. J., Malavolta L., Kunovac V., Armstrong D. J., **Attia M.**, Barros S. C. C. et al. Rossiter–McLaughlin detection of the 9-month period transiting exoplanet HIP41378 d. *Astronomy & Astrophysics* 668, A172 (2022).

7. Almenara J. M., Bonfils X., Otegi J. F., **Attia M.**, Turbet M., Astudillo-Defru N., Collins K. A., Polanski A. S., Bourrier V., Hellier C. et al. GJ 3090 b: one of the most favourable mini-Neptune for atmospheric characterisation. *Astronomy & Astrophysics* 665, A91 (2022).

6. Bourrier V., Zapatero Osorio M. R., Allart R., **Attia M.**, Cretignier M., Dumusque X., Lovis C., Adibekyan V., Borsa F., Figueira P. et al. The polar orbit of the warm Neptune GJ 436b seen with VLT/ESPRESSO. *Astronomy & Astrophysics* 663, A160 (2022).

5. Katz H., Martin-Alvarez S., Rosdahl J., Kimm T., Blaizot J., Haehnelt M. G., Michel-Dansac L., Garel T., Oñorbe J., Devriendt J. et al. Introducing SPHINX-MHD: the impact of primordial magnetic fields on the first galaxies, reionization, and the global 21-cm signal. *Monthly Notices of the Royal Astronomical Society* 507, 1254 (2021).

4. Pezzotti C., **Attia M.**, Eggenberger P., Buldgen G., Bourrier V. The key impact of the host star's rotational history on the evolution of TOI-849b. *Astronomy & Astrophysics* 654, L5 (2021).

3. **Attia M.**, Teyssier R., Katz H., Kimm T., Martin-Alvarez S., Ocvirk P., Rosdahl J. Cosmological magnetogenesis: the Biermann battery during the Epoch of reionization. *Monthly Notices of the Royal Astronomical Society* 504, 2346 (2021).

2. **Attia M.**, Bourrier V., Eggenberger P., Mordasini C., Beust H., Ehrenreich D. The JADE code: Coupling secular exoplanetary dynamics and photo-evaporation. *Astronomy & Astrophysics* 647, A40 (2021).

1. Seidel J. V., Ehrenreich D., Bourrier V., Allart R., **Attia M.**, Hoeijmakers H. J., Lendl M., Linder E., Wyttenbach A., Astudillo-Defru N. et al. Hot Exoplanet Atmospheres Resolved with Transit Spectroscopy (HEARTS): V. Detection of sodium on the bloated super-Neptune WASP-166b. *Astronomy & Astrophysics* 641, L7 (2020).

CONTRIBUTIONS IN INTERNATIONAL CONFERENCES

- 2026 November • Exobiologie Jeunes Chercheur·ses (EJC), Paris (France) • **invited talk**
- 2026 September • Europlanet Science Congress (EPSC), The Hague (Netherlands) • oral presentation • **session convener**
- 2026 June • Journées de la Société Française d'Astronomie et d'Astrophysique (SF2A), Grenoble (France) • poster
- 2026 April • Layers of Understanding, Heidelberg (Germany) • oral presentation
- 2026 January • Rocky Worlds 4, Groningen (Netherlands) • **LOC**
- 2025 October • Swiss Society for Astronomy and Astrophysics (SSAA) General Assembly, Locarno (Switzerland) • oral presentation
- 2024 June • Exoplanets 5, Leiden (Netherlands) • poster
- 2023 July • Towards other Earths III, Porto (Portugal) • oral presentation
- 2023 April • Protostars and Planets VII, Kyoto (Japan) • poster
- 2022 June • European Astronomical Society (EAS) Annual Meeting, Valencia (Spain) • poster
- 2022 June • Journées de la Société Française d'Astronomie et d'Astrophysique (SF2A), Besançon (France) • oral presentation
- 2022 April • PlanetS General Assembly, Grindelwald (Switzerland) • oral presentation
- 2021 December • ExoSystèmes II, Toulouse (France) • oral presentation
- 2021 September • Europlanet Science Congress (EPSC), virtual • oral presentation (awarded the virtual Outstanding Presentation Contest)
- 2021 June • Programme National de Planétologie (PNP), organized by SF2A, virtual • oral presentation

COLLOQUIUM AND SEMINAR TALKS

- 2027 January • Institut de Planétologie et d'Astrophysique de Grenoble, Université Grenoble-Alpes • Grenoble (France)
- 2026 November • Laboratoire de Planétologie et Géosciences, Nantes Université • Nantes (France)
- 2026 October • Institute of Astronomy, University of Cambridge • Cambridge (UK)
- 2025 July • Kapteyn Astronomical Institute, University of Groningen • Groningen (Netherlands)
- 2023 November • Centre for Exoplanets and Habitability, University of Warwick • Coventry (UK)
- 2023 April • National Astronomical Observatory of Japan • Tokyo (Japan)
- 2022 April • Institut d'Astrophysique de Paris, Sorbonne Université • Paris (France)
- 2022 March • Observatoire de Genève, Université de Genève • Versoix (Switzerland)
- 2021 November • Institute for Computational Science, University of Zurich • Zurich (Switzerland)
- 2021 November • Institut de Recherche en Astrophysique et Planétologie, Université de Toulouse • Toulouse (France)
- 2021 May • Institut de Planétologie et d'Astrophysique de Grenoble, Université Grenoble-Alpes • Grenoble (France)
- 2021 April • Observatoire de Genève, Université de Genève • Versoix (Switzerland)

PUBLISHED OPEN-SOURCE CODES

- PALEOS: a comprehensive Python package for computing equations of state and thermodynamic properties of planetary materials under extreme pressure–temperature conditions relevant to planetary interiors.
- PROTEUS: a modular Python framework for the coupled interior–atmosphere–orbit evolution of rocky and sub-Neptune planets, developed as a team effort by the whole Forming Worlds Lab and its collaborators, with Mara as a maintainer and major contributor.
- ZEPHYRUS: the atmospheric escape module of PROTEUS, whose development Mara now leads and has rebuilt around a regime dispatcher that selects among six escape regimes and splits the outflow over species by fractionation.
- TripleBHMergerPredictor: a Python-based neural network for predicting mergers in hierarchical triple black hole systems trained on ~15M secular simulations.
- JADE: a sophisticated Python framework for simulating the coupled evolution of exoplanetary atmospheres and orbital dynamics in hierarchical three-body systems; used in 12 studies to date.

OBSERVATIONS

- Former regular observer with the SOPHIE instrument on the 193-cm telescope at Observatoire de Haute-Provence, France.
- Former regular observer on the EULER telescope in La Silla, Chile.
- PI of 1 observing program on ESPRESSO (2 nights), 1 observing program on HARPS (1 night), 1 observing program on SOPHIE (8.3 nights), and 1 observing program on GIARPS (2 nights).
- Co-I of 4 observing programs on ESPRESSO, 1 observing program on HARPS, 1 observing program on NEID, 1 observing program on NIRSPEC, and 1 observing program on CHEOPS.

SUPERVISION

- 2026 April – 2026 July • co-supervised Viesturs Strelčs (University of Groningen) for a BSc thesis on diffusion-limited atmospheric escape.
- 2026 April – 2026 July • co-supervised Malina Ovesen (University of Groningen) for a BSc thesis on radiation-recombination-limited atmospheric escape.
- 2026 April – 2026 August • co-supervised Renske Beuker (University of Groningen) for a BSc thesis on impact-driven atmospheric escape.
- 2026 April – 2026 July • co-supervised Ioana Balint (University of Groningen) for a BSc thesis on Jeans atmospheric escape.
- 2025 September – 2026 December • co-supervised Ema Jungová (University of Groningen) for a MSc thesis on sub-Neptune ocean formation.
- 2023 May – 2023 June • co-supervised Léonie Hoerner (Université de Lorraine) for a MSc internship on modeling exoplanetary water atmospheres.
- 2022 April – 2022 August • co-supervised Théo Vrignaud (École polytechnique, Paris) for a MSc thesis on the analysis of the Rossiter–McLaughlin effect of ultra-short-period rocky exoplanets.
- 2022 February – 2022 June • supervised Leon Kawang Kwok (University of Geneva) for a MSc project on exoplanetary dynamical tides.
- 2021 November – 2022 March • supervised Aymeric Carchereux (University of Geneva) for a BSc project on exoplanetary radius pulsations induced by the Kozai–Lidov mechanism.
- 2021 September – 2022 January • co-supervised Nathan Monnet (École Polytechnique Fédérale de Lausanne) for a MSc project on the analysis of the Rossiter–McLaughlin effect of the Moon.

TEACHING

- 2026 Fall & 2025 Fall • invited lecturer for *Planetary Systems* (BSc, University of Groningen)
- 2026 Fall • invited lecturer for *Astrobiology* (BSc, University of Groningen)
- 2026 Spring • invited lecturer for *Exoplanets* (MSc, University of Groningen)
- 2023 Spring • teaching assistant for *Planetary Atmospheres* and *Planetary Formation and Evolution* (MSc, University of Geneva)
- 2022 Spring • teaching assistant for *Planetary Atmospheres* (MSc, University of Geneva)

PROFESSIONAL SERVICE

- Referee for the journals *Astronomy & Astrophysics*, *The Astrophysical Journal*, and *The Astrophysical Journal Letters*.

OUTREACH

- Talk on astronomy and exoplanet research for prospective students at the Open Day of the Faculty of Science and Engineering, University of Groningen (Groningen, Netherlands).
- Public lecture on exoplanet science at the "Eureka!" science festival (Groningen, Netherlands).
- Public talk introducing exoplanet science in a local bar, at one of the "Astronomy on Tap Groningen" evenings (Groningen, Netherlands).
- Public talk on exoplanet science at the "Nacht van de Nacht 2025" light-pollution awareness event (Groningen, Netherlands).

- Interview in the Meridiana journal (Switzerland) about the results of Rodrigues et al. (2025, *A&A*, 695, A237).
- Press release on the results of Rodrigues et al. (2025, *A&A*, 695, A237), issued by the University of Geneva.
- Poster on the hot Neptune desert, designed with a student from the École supérieure de bande dessinée et d'illustration de Genève for the "Exoplanètes : Art, Science & Fiction" exhibition, which paired researchers from the Exoplanets group of the Geneva Observatory with illustration students.
- Question-and-answer sessions with the general public on astrophysics-related topics, as part of the "Astro-coffees" project organized by the Geneva Observatory and the Geneva Museum of Sciences.
- Press releases on the results of Bourrier, Attia et al. (2023, *A&A*, 669, A63), issued by the University of Geneva and ESO.

SOCIAL

- Active member of the Diversity, Equity and Inclusion committee of the Geneva Observatory. Organized several events to promote diversity and inclusion at the Observatory.
- Strongly involved locally in advocating for LGBTQ+ rights and helping precarious transgender communities in Geneva.

SPOKEN LANGUAGES

French (native) • Arabic (native) • English (fluent) • German (intermediate) • Dutch (beginner)